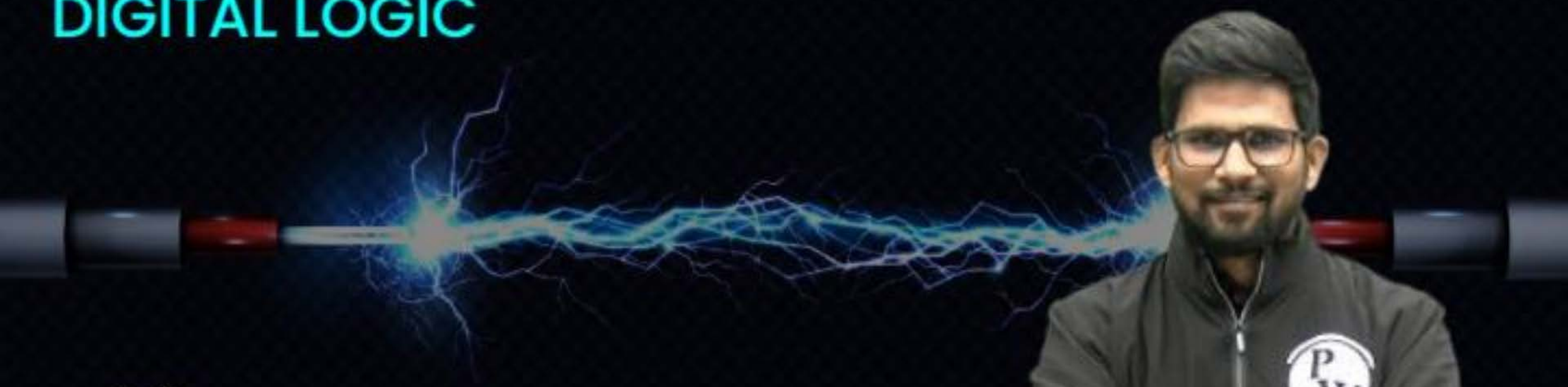
COMPUTER SCIENCE & IT

DIGITAL LOGIC



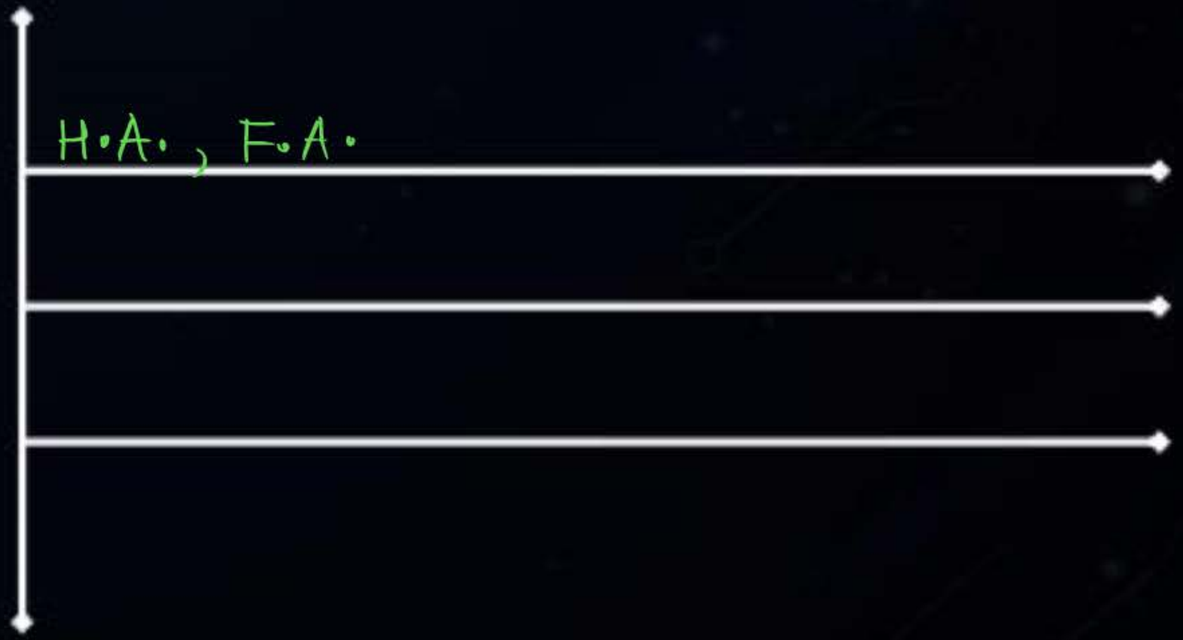
Lecture No. 04

Combinational Circuit



By- Chandan Gupta Sir

Recap of Previous Lecture





H.S., F.S.

- To add two n-bit numbers → How many HA's and OR gates required ?

n F·A

$2n$ H·A

n OR

X

$\Rightarrow$ 1 H·A + $(n-1)$ F·A

1 H·A + $2(n-1)$ H·A + $(n-1)$ OR gate

$(2n-1)$ H·A + $(n-1)$ OR gate

$a_3\ a_2\ a_1\ a_0$

$b_3\ b_2\ b_1\ b_0$

$c_3\ c_2\ c_1\ 0$

$c_4\ s_3\ s_2\ s_1\ s_0$

Question 1

We have to add two 4-bit numbers A and B using parallel adder. Sum delay and carry delay of each F.A. is 3-nsec and 2-nsec respectively, then time required to perform complete addition

(a) 9 nsec
(b) 8 nsec
(c) 12 nsec
(d) 20 nsec

$n=4 \rightarrow$ $t_s = 3\,ns$

$t_c = 2\,nsec$

$T_d = (n-1)t_c + \max[t_c, t_s]$

$= 3t_c + t_s$

$= 6 + 3 = 9\,nsec.$

Question 2

If in above question sum delay is 2 nsec and carry delay is 3 nsec then time require to perform complete addition will be 12 nsec.

$t_s = 2\,ns$

$t_c = 3\,nsec$

$T_d = (n-1)t_c + \max(t_c, t_s)$

$= 3t_c + t_c$

$= 4t_c$

$= 4 \times 3 = 12\,nsec$

Question 3

We have two 100 bit numbers added using parallel adder. Sum and carry delay of each F.A. used are 2.24 nsec and 1.24 nsec respectively, then number of addition performed by this parallel adder in 1 sec

$__8__ \times 10^6$/sec.

$\Rightarrow$ $t_s = 2.24$ nsec

$t_c = 1.24$ nsec

$n = 100$

time for one addition $T_d = (n-1)t_c + \max[t_c, t_s]$

$T_d = 99t_c + t_s$

$= 99 \times 1.24 + 2.24$

$= 124 + 1$ $(1.24 + 1)$

$= 125$ nsec

$\Rightarrow N = \dfrac{1 \text{ sec}}{125 \text{ nsec}}$

$= \dfrac{10^9}{125}$

$8 \times 10^6 = \dfrac{10^3}{125} \times 10^6$

Question 4

To add two 4-bit numbers, minimum number of H.A.s require is m and minimum number of OR gates require is n then value of $(m + n)$ is

10 .

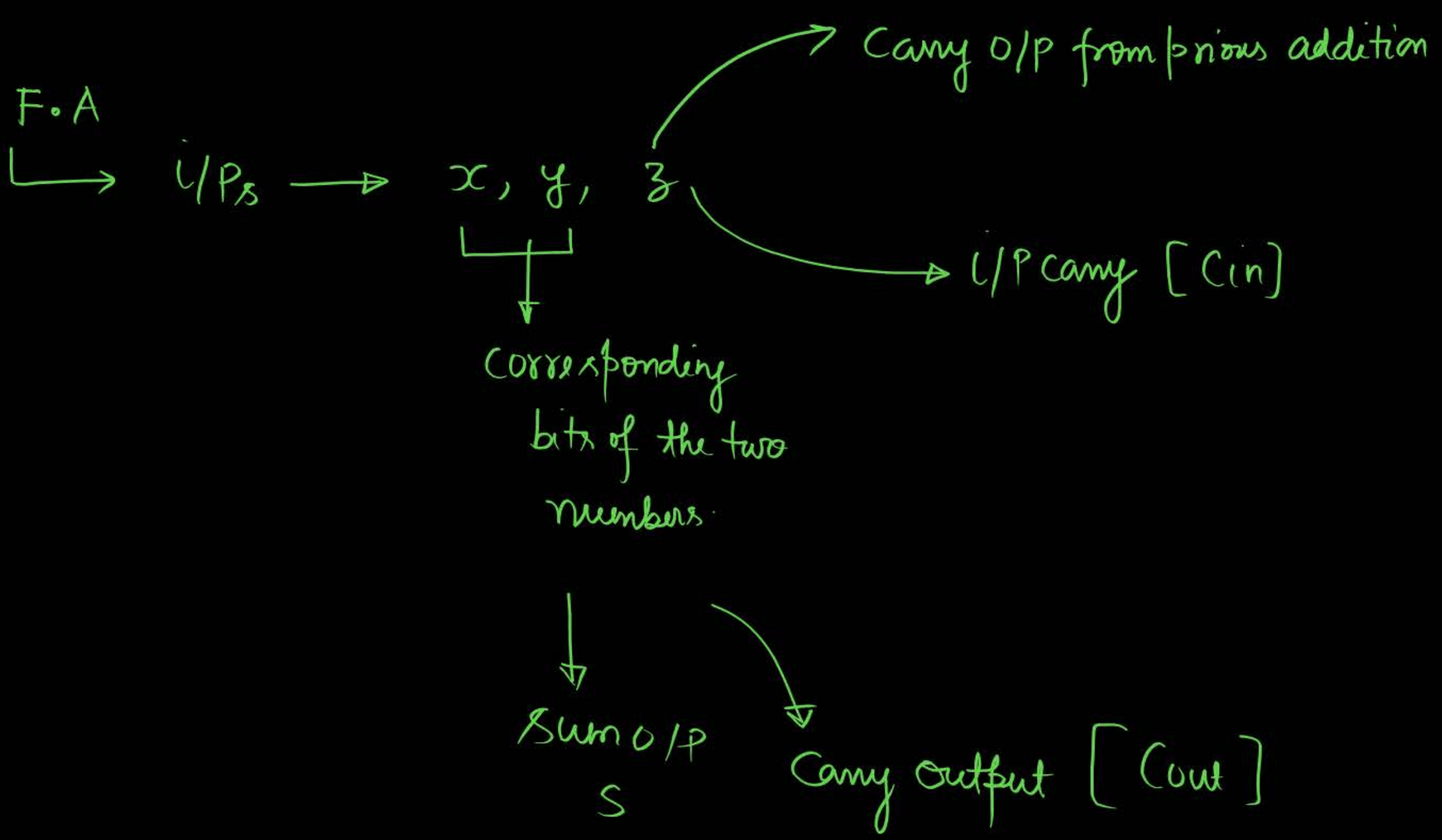
1 H.A + 3 F.A.

= 1 HA + 6 H.A + 3 OR

= 7 H.A + 3 OR

$m = 7$

$n = 3$

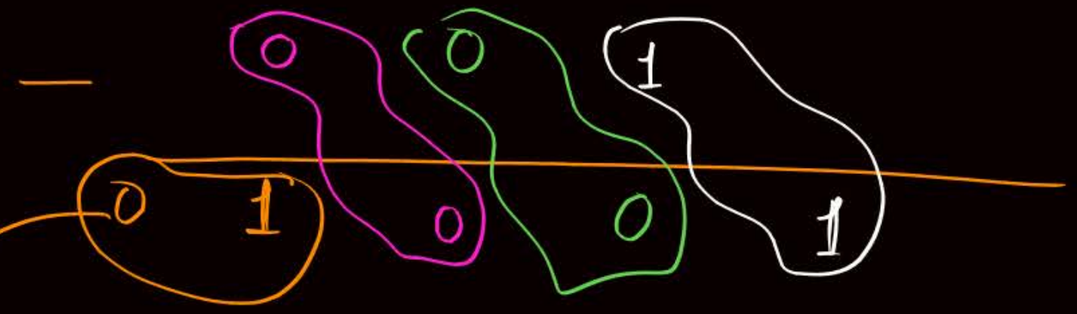

F.A
i/Ps
x, y, z
Carry o/p from previous addition
i/P carry [Cin]
Corresponding bits of the two numbers
Sum o/p S
Carry output [Cout]

- $A = (1110)_2 = (14)_{10}$

  $B = (0101)_2 = (5)_{10}$

$(A-B) \Rightarrow$

$$\begin{array}{cccc} 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ \hline 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 \end{array}$$



it is signifying that result is +Ve ⟶ result is $(1001)_2 = (9)_{10}$

$$\begin{array}{r} (786)_{10} \\ -\ 298 \\ \hline 488 \end{array}$$

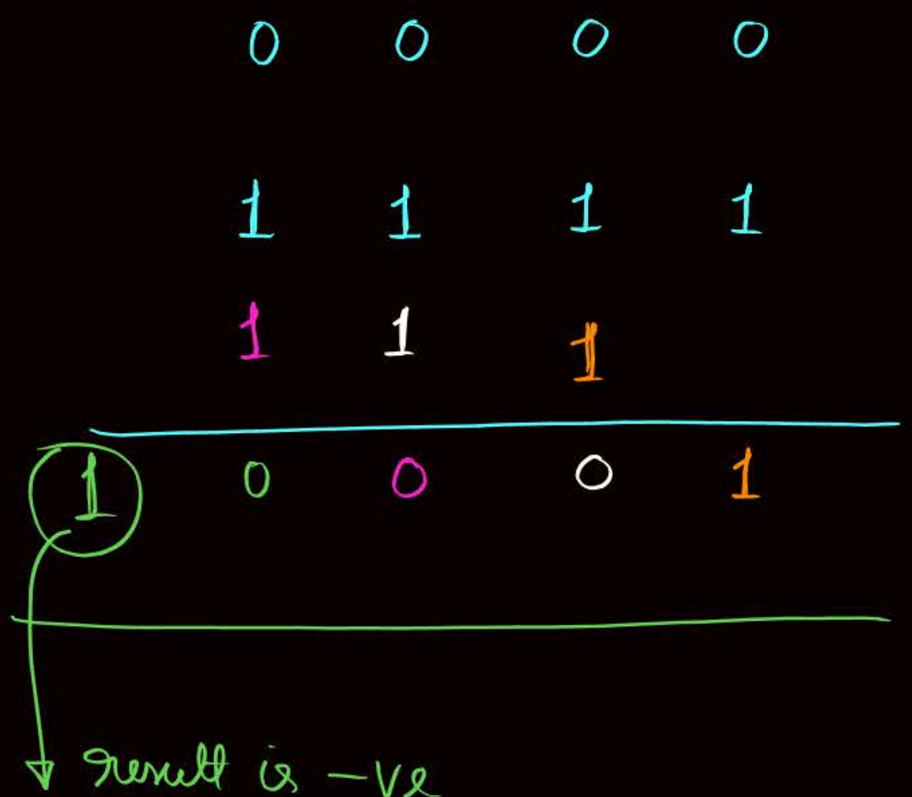
the result is +Ve.

B–A ⇒

	0	1	0	1
	1	1	1	0
−	1	1	0	
1	0	1	1	1 ✓

this is signifying that result is –ve.

the result is $-\overline{[2's \text{ complement of } 0111]} = -[1001] = -[9]$



the result is $\Rightarrow -[2\text{'s } 0001] = -[1111]_2 = -(15)_{10}$

- $A = a_3\ a_2\ a_1\ a_0$

— $B = b_3\ b_2\ b_1\ b_0$

$B_3\ B_2\ B_1$

Result $B_4\ D_3\ D_2\ D_1\ D_0$

Half Subtractor

- What is Half subtractor and what are input lines and what are output lines ?

2- i/p lines ⟹ (x, y) ⟶ the corresponding bits of the two numbers being subtracted

2- o/p lines ⟶ Difference o/p D

Borrow o/p B.

$(x-y)$

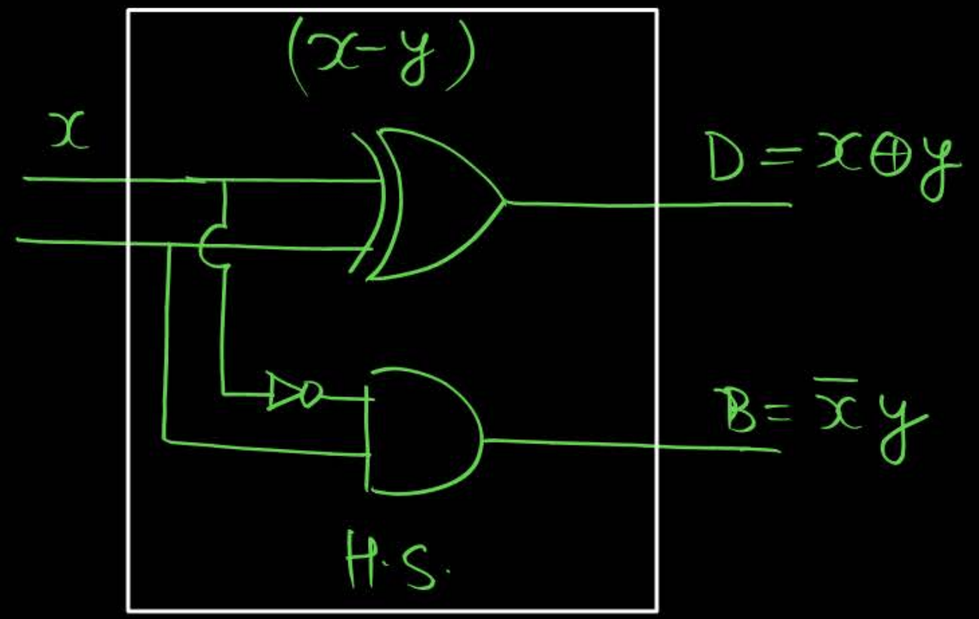
	Input lines		Output lines	
	x	y	D-o/p	B-o/p
0	0	0	0	0
1	0	1	1	1
2	1	0	1	0
3	1	1	0	0

$$D(x,y) = \Sigma(1,2) = \bar{x}y + x\bar{y} = x \oplus y$$

↳ non-self dual

$$B(x,y) = \Sigma(1) = \bar{x} \cdot y$$

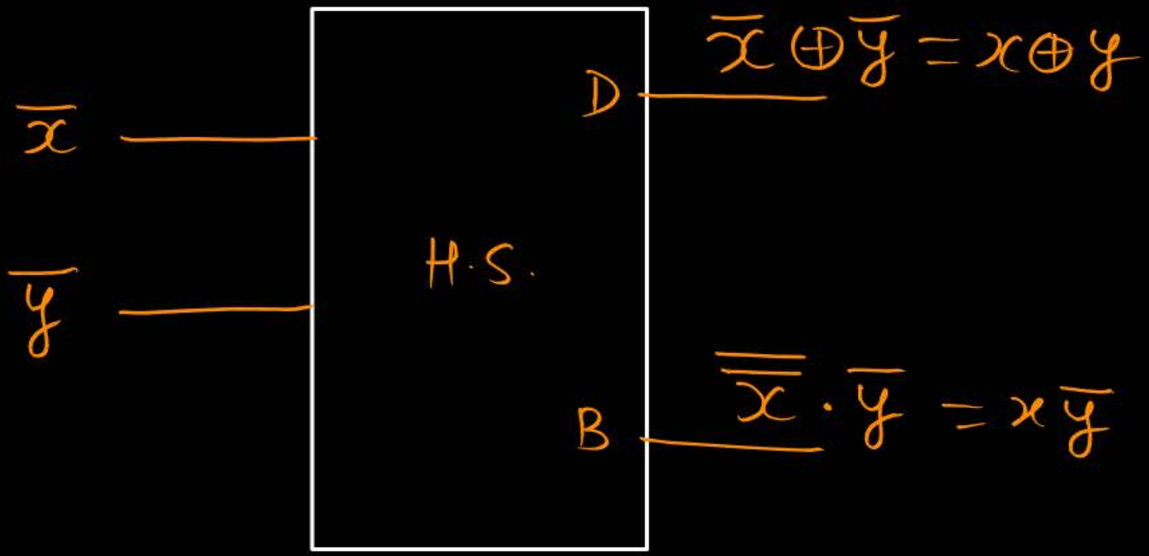
↳ non-self dual



$y-x \longrightarrow D = y \oplus x = x \oplus y$

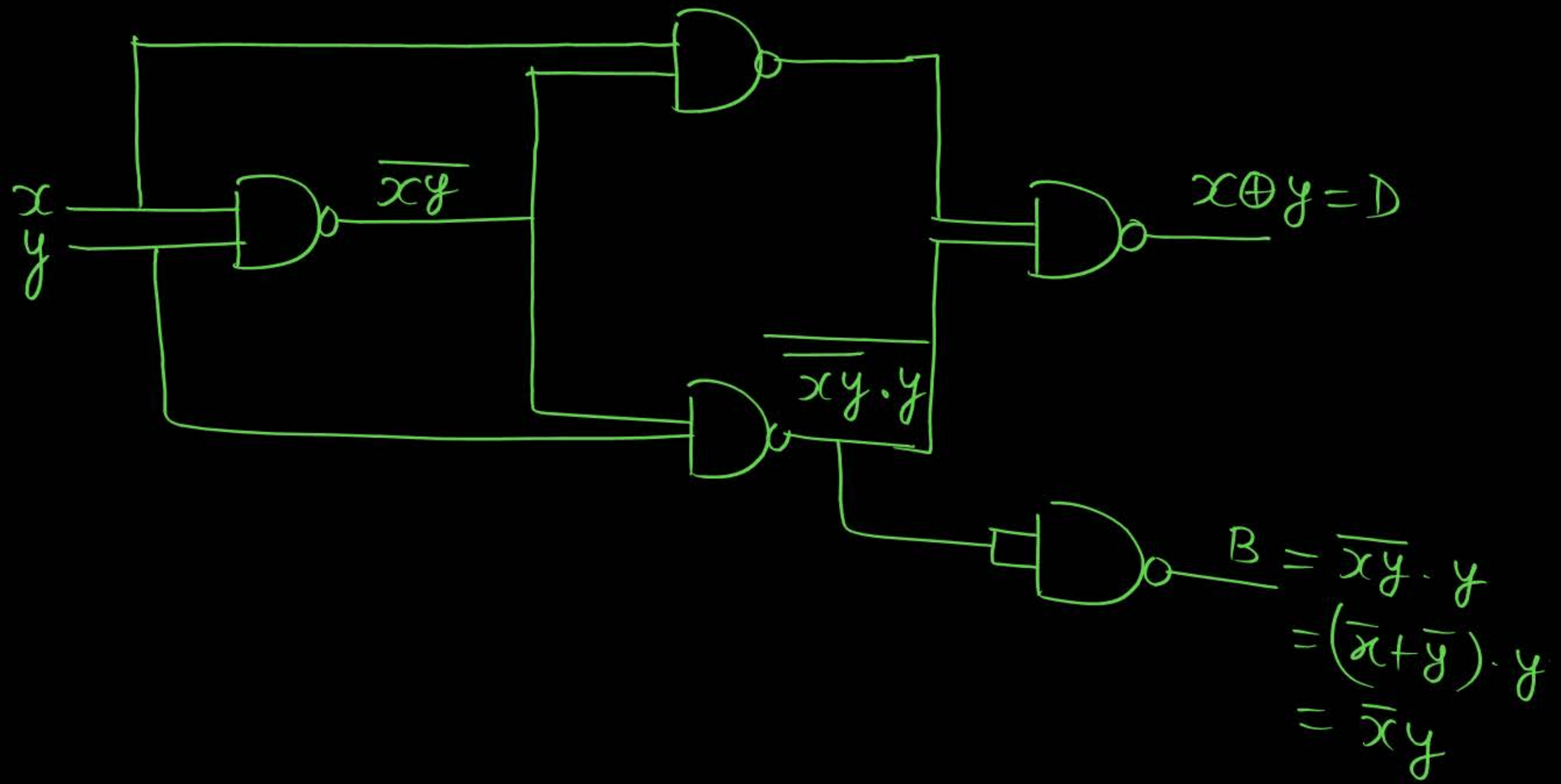
↳ $B = \bar{y} \cdot x$

Implementation using gates :

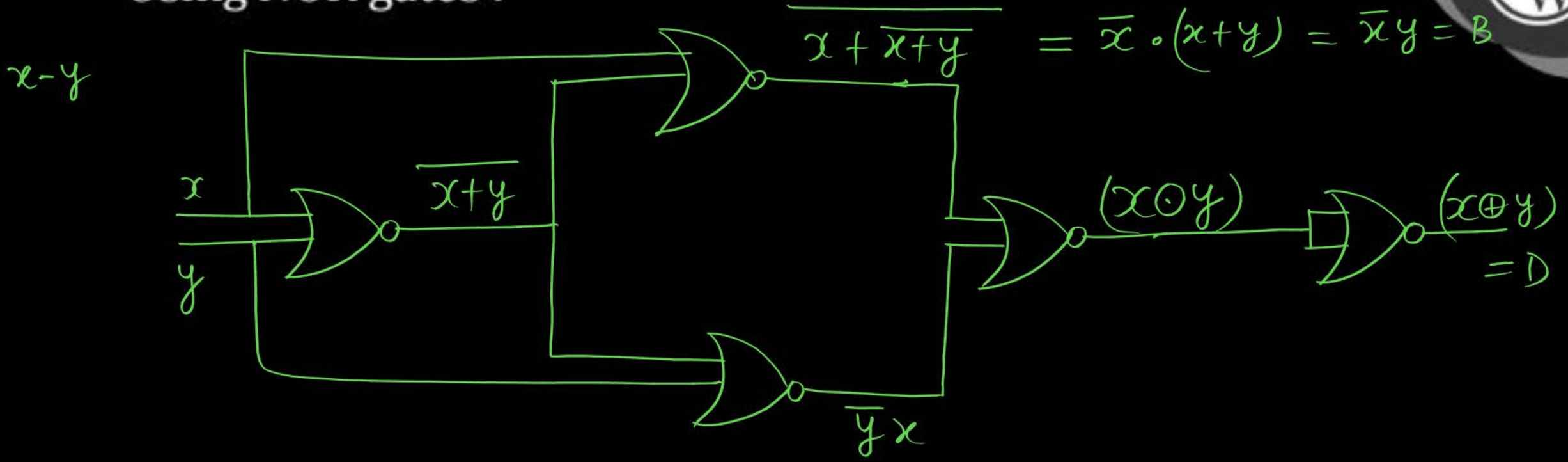


- Using NAND gates :

$x-y$



- Using NOR gates :



⇒ To implement H.S. we require 5 (2-i/p) NAND gates or 5 (2-i/p) NOR gates.

Full Subtractor

- What is full subtractor and what are input lines and what are output lines ?

3-i/p lines ⟶ (x, y, z)

z → borrow o/p from the previous subtraction.

x, y → corresponding bits of the two no.s being subtracted.

$(x - y - z)$

2-o/p lines ⟶ Difference o/p D

Borrow B

x - y - z

	Input lines			Output lines	
	x	y	z	D-o/p	B-o/p
0	0	0	0	0	0
1	0	0	1	1	1
2	0	1	0	1	1
3	0	1	1	0	1
4	1	0	0	1	0
5	1	0	1	0	0
6	1	1	0	0	0
7	1	1	1	1	1

$D(x, y, z) = \Sigma(1,2,4,7) = (x \oplus y \oplus z)$

↳ Self dual boolean function

$B(x, y, z) = \Sigma(1,2,3,7)$

↳ Self dual boolean function

$= \bar{x}\bar{y}z + \bar{x}y\bar{z} + \bar{x}yz + xyz$

$= (\bar{x}y + \bar{x}z + yz)$

$= \bar{x}\bar{y}z + \bar{x}y\bar{z} + \bar{x}yz + xyz = \bar{x}y + z(x \odot y)$

Implementation using gates :

$x+y+z$

$xy+yz+zx \longrightarrow xy+(x \oplus y)z$

$x-y-z \longrightarrow \bar{x}y+yz+\bar{x}z$

$$\bar{x}y+(\bar{x} \oplus y)z = \bar{x}y+(x \odot y)z$$

$x+y+z \longrightarrow xy+yz+xz$

$x \cdot y + (x \oplus y) \cdot z$

$(y+z+x) \longrightarrow yz + (y \oplus z) \cdot x$

$(z+x+y) \longrightarrow x \cdot z + (x \oplus z) \cdot y$

$x-y-z \Rightarrow \bar{x}y + \bar{x}z + yz$

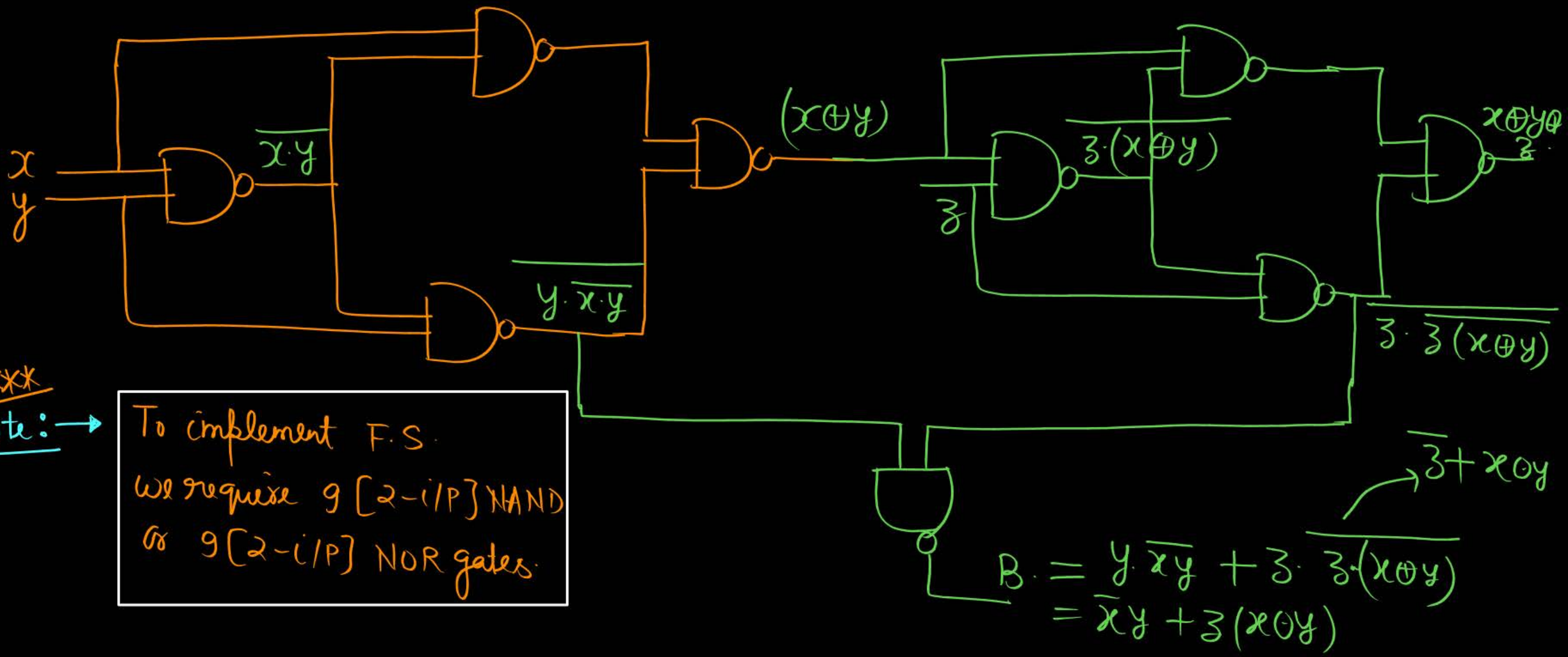
$\bar{x}y + (\bar{x} \oplus y) \cdot z$

$x-z-y = \bar{x}z + (\bar{x} \oplus z) \cdot y$

$= \bar{x}z + (x \odot z) \cdot y$

$(x-y-z)$

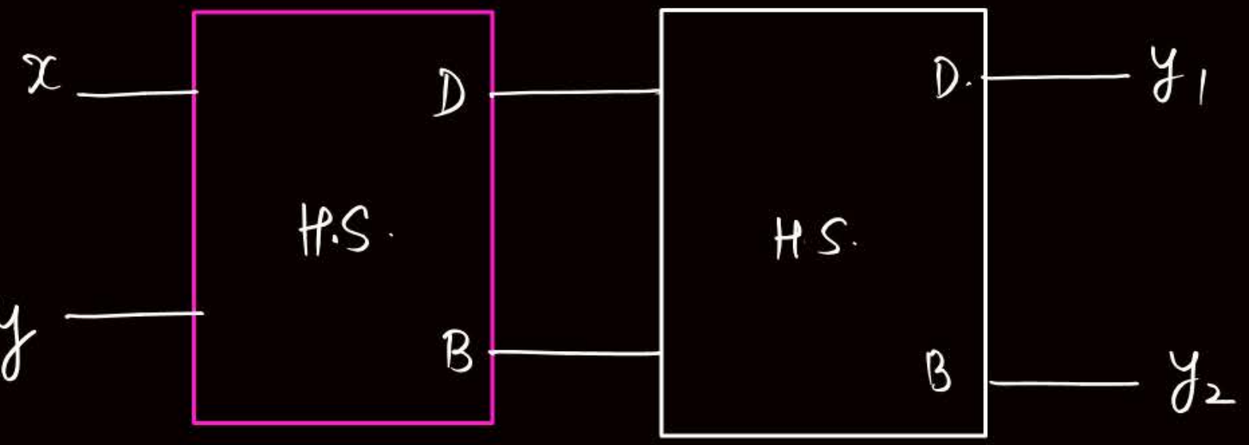
- Implementation using NAND and NOR gates : → H.W.



***
Note: → To implement F.S. we require 9 [2-i/p] NAND or 9 [2-i/p] NOR gates.

H.W.

Q.



Q. $A = (1010)_2$

$B = (0010)_2$

perform (A−B) & (B−A) and find out result.

Q. $A = (0111)_2$ What is the result of (A−B) ?

$B = (1011)_2$



2 Minute Summary

H.S. & F.S.

Thank you

GW

Soldiers !

